MIL-T-21669(NOrd)

12 November 1958
SUPERSEDING
NAVORD OS 5879A
18 August 1952

MILITARY SPECIFICATION

TEST SET MARK 25 MOD 1

1. SCOPE

1.1 This specification establishes the requirements for the procurement of Test Set Mark 25 Mod 1. This test set checks continuity and resistance, and measures current in Sterilizer Mechanisms SD 4 Mod 1 and 5 Mod 0.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	Boxes, Fiber.

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-B-131	Barrier Material: Water vaporproof, Flexible.
MIL-V-173	Varnish, Moisture and Fungus resistant for the Treatment of Communications, Electronic and Associated Electrical Equipment.
MIL-P-17555(SHIPS)	Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts.

FSC 6625

Source: https://assist.dla.mil -- Downloaded: 20260-19T10:16Z
Check the source to verify that this is the current version before use.

MIL-S-6872	Soldering Processes, General Specifications for.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWINGS

BUREAU OF ORDNANCE

LD 130779	Test Set Mark 25 Mod 1
LD 284866	Preparation for Delivery For Test Set Cable Assemblies.
1476194	Marking Instructions for Storage and Shipping Containers.

And all documents listed in the above LD's and drawings.

PUBLICATIONS

BUREAU OF ORDNANCE

NAVORD OP 400	General Instructions for the Design, Manufacture and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements.

3.1.1 Test Set Mark 25 Mod 1 shall be manufactured in accordance with LD 130779 and all documents listed thereon.

3.1.2 The requirements and procedures established in OP 400 shall be complied with as applicable.

Source: https://assist.dla.mil -- Downloaded: 20260-19T10:16Z
Check the source to verify that this is the current version before use.

3.1.3 Preproduction sample. - Unless otherwise specified in the contract or order, a preproduction sample shall be manufactured by the same procedures and processes and at the same location as proposed for full production. This sample will be tested for the purpose of determining compliance with the requirements of this specification, the drawings, and the contract. See 4.1.2.

3.1.4 Interchangeability. - Component parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in strict agreement with their applicable drawings, shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required and changes in dimensions are necessary, no changes in the parts as delineated on applicable drawings shall be made without prior approval of the Bureau of Ordnance.

3.2 Moisture - and fungus-resistant treatment. - Materials and parts of this test set which are not moisture and fungus resistant shall be treated, before assembly, with a varnish conforming to Military Specification MIL-V-173.

3.3 Requests for deviation. - Requests for deviation from this specification, applicable drawings, specifications, publications, materials and processes specified herein, shall be as set forth in OP 400.

3.4 Conflicting requirements. - Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein, shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.5 Soldering. - Unless otherwise specified, soldering processes shall conform to the requirements specified in Military Specification MIL-S-6872. Only non-corrosive and non-volatile fluxes shall be used. Rosin shall be used when soldering wires to contacts and terminals. The use of rosin core solder is permitted. Electrical connections shall be mechanically secure and electrically continuous before soldering. Sufficient heat shall be applied during the soldering process to avoid "cold" solder joints.

Source: https://assist.dla.mil -- Downloaded: 20260-19T10:16Z
Check the source to verify that this is the current version before use.

3.6 Wiring. - All internal wiring shall be cabled wherever possible.

3.7 Performance requirements and product characteristics. - The test set covered by this specification shall meet the following performance requirements and product characteristics.

3.7.1 Insulation resistance. - The insulation resistance between electrical circuits and between each circuit and the chassis shall be not less than 50 megohms when tested as specified in 4.5.3.

3.7.2 Ohmmeter. - The test set meter shall indicate 1000, 4000, and 10,000 ohms (resistor values) ± 2 percent when connected and tested as specified in 4.5.4.

3.7.3 Ammeter. - The test set meter shall indicate the same as an external ammeter, within ± 2 percent, when connected and tested as specified in 4.5.5.

3.8 Workmanship and cleanliness. - The test set shall be free from dust, dirt, metal filings, moisture, solder splashes, chips, and other foreign matter. Necessary precautions shall be taken during manufacture and assembly to insure these conditions.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS.

4.1 Government inspection. - The Government inspector will make such inspections as are necessary to determine that all components and assemblies are in accordance with the requirements of the contract, pertinent specifications and drawings. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in OP 400.

4.1.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection will be in accordance with the provisions of MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 20260-19T10:16Z
Check the source to verify that this is the current version before use.

4.1.2 Preproduction sample. - Unless otherwise specified for each contract or order, a representative sample of one test set shall be manufactured using the methods and procedures proposed for full production. The Government inspector shall perform, witness, or participate in such inspections and tests on the sample as are required for full production. Upon conditional acceptance by the inspector, the preproduction sample shall be forwarded, together with a report of inspection, to an activity designated in the contract or by the procuring agency, for the purpose of determining compliance with the requirements of the contract, specifications and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of the tests on preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.2.1 Preproduction sample for a subsequent contract. - The necessity for a preproduction sample will be determined by the procuring agency when production under a new contract by the same contractor at the same location follows the manufacture of satisfactory material as covered by this document.

4.1.3 Lot. - As applied to the Government inspection of units of product, the term "lot" shall mean "inspection lot", i.e. a collection of units of the product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" shall be as determined by the Government inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Contractor's inspection. - Contractor's inspection shall be conducted in accordance with provisions of "Inspection by Contractor" as contained in OP 400.

4.3 Classification of tests. - The inspection and testing of mechanisms covered by this document shall be classified as follows:

(a) acceptance tests

(b) preproduction tests

Source: https://assist.dla.mil -- Downloaded: 202640-19T10:16Z
Check the source to verify that this is the current version before use.

4.3.1 Acceptance tests. - Acceptance tests for acceptance of the test set covered by this specification shall consist of tests to determine compliance with all requirements specified in this document.

4.3.2 Preproduction tests. - Preproduction tests are those performed on samples submitted in accordance with 4.1.2 to determine that the lot and production meet all the requirements of this specification. Acceptance shall be based on no defects in a sample.

4.4 Test equipment.

4.4.1 Test and inspection equipment and facilities. - Unless otherwise specified, the manufacturer shall furnish and maintain all necessary test equipment, facilities and personnel for performing all acceptance tests under the supervision of the Government inspector. The test equipment shall be adequate in quantity, and unless definite requirements are specified, they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.4.1.1 Equipment required for performing these tests is as follows:

(a) One (1) 1.5-volt battery, flashlight type.

(b) One (1) resistor, 1000 ohms ± 1 percent, 1-watt.

(c) One (1) resistor, 4000 ohms ± 1 percent, 1-watt.

(d) One (1) resistor, 10,000 ohms ± 1 percent, 1-watt.

(e) One (1) potentiometer, 3000 ohms, 1-watt.

(f) One (1) milliammeter, 0-10 milliamperes range, accuracy within ± 2 percent.

(g) One (1) megohmmeter 500-volts d-c. Range 10.0-100 megohms, accuracy within ± 10 percent.

Source: https://assist.dla.mil -- Downloaded: 20260-19T10:16Z
Check the source to verify that this is the current version before use.

4.5 Tests.

4.5.1 Test conditions. - Unless otherwise specified, tests shall be performed at 80° ± 10° F and ambient relative humidity.

4.5.2 Visual inspection. - The test set and its component parts shall be inspected to verify that the materials, design, construction, physical dimensions, soldering and workmanship are in accordance with this document and the drawings.

4.5.3 Insulation resistance. - The test set shall be subjected to a conditioning period of 24 hours at a temperature of 77° ± 2° F, and a relative humidity of 75 ± 5 percent. A test potential of 500 ± 50 volts d-c is then applied, using megohmmeter (item 4.4.1.1 (g)), as specified in 3.7.1. The insulation resistance shall be not less than 50 megohms.

4.5.4 Ohmmeter tests.

(a) Set selector switch at OHMS.

(b) Jumper CURRENT-OHMS binding posts. It shall be possible to cause test set meter to indicate zero ohms by turning control below meter.

(c) With meter indicating zero, remove jumper. DO NOT TOUCH CONTROL AGAIN DURING THIS TEST.

(d) Connect the 1000-ohm precision resistor (item 4.4.1.1 (b)) across CURRENT-OHMS binding posts. Test set meter shall indicate 1000 ± 20 ohms.

(e) Remove resistor.

(f) Repeat steps (d) and (e) using 4000-ohm (item 4.4.1.1 (c)) and 10,000 ohm (item 4.4.1.1 (d)) resistors. In each case, test set meter shall indicate resistor value ±2 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:16Z
Check the source to verify that this is the current version before use.

4.5.5 Ammeter tests.

(a) Set 3000-ohm potentiometer (item 4.4.1.1 (e)) so that its total resistance is in effect.

(b) Observing polarity, connect 1.5-volt battery (item 4.4.1.1 (a)), potentiometer (with one end open), and external milliammeter (item 4.4.1.1 (f)) in series across the CURRENT-OHMS binding posts.

(c) Set test set selector switch at 1 MA. Test set meter shall indicate same as external milliammeter within ± 2 percent.

(d) Vary external potentiometer setting (making sure both meter pointers stay on scale) until external meter indicates approximately twice the value obtained in step (c). Test set meter shall indicate same as external milliammeter within ± 2 percent.

(e) Set test set selector switch at 3 MA. Test set meter shall indicate same as external milliammeter within ± 2 percent.

(f) Vary external potentiometer setting (making sure both meter pointers stay on scale) until external meter indicates approximately twice the value obtained in step (e). Test set meter shall indicate same as external milliammeter within ± 2 percent.

(g) Set test set selector switch at 10 MA. Test set meter shall indicate same as external milliammeter within ± 2 percent.

(h) Vary external potentiometer setting (making sure both meter pointers stay on scale) until external meter indicates approximately twice the value obtained in step (g). Test set meter shall indicate same as external milliammeter within ± 2 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:16Z
Check the source to verify that this is the current version before use.

(i) Set test set selector switch at 30 MA. Test set meter shall indicate same as external milliammeter within ± 2 percent.

(j) Vary external potentiometer setting (making sure both meter pointers stay on scale) until external meter indicates approximately twice the value obtained in step (i). Test set meter shall indicate same as external milliammeter within ± 2 percent.

(k) Remove milliammeter, potentiometer, and the battery.

4.5.6 Acceptance criteria. - Failure of any test set to meet any of the requirements and tests of this specification shall result in rejection of the test set. If the number of failing test sets exceeds the number permitted by the associated criteria, the lot represented will be rejected.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. - In addition to the following, Level A packaging shall be in accordance with Specification MIL-P-17555.

5.1.1.1 Cleaning, drying, and preservative application. - (Not applicable)

5.1.1.2 Unit packaging. - Prior to unit packaging, all test set cables shall be packaged in accordance with Bureau of Ordnance LD No. 284866. Each test set shall be packaged in a Type I, Class 1, Style RSC corrugated fiberboard box, having a minimum Mullen test of 275 pounds, conforming to the requirements of Federal Specification PPP-B-636. Sufficient fillers and folded corner pads, fabricated from the same box material, shall be utilized within the carton to provide a tight pack and to prevent direct contact between protrusions of the test set case and the walls of the fiberboard box.

Source: https://assist.dla.mil -- Downloaded: 20260-19T10:16Z
Check the source to verify that this is the current version before use.

5.1.1.2.1 The unit package, prepared as described above, shall be further packaged Method IA-14 conforming to the requirements of Military Specification MIL-P-116. The inner carton with all corners blunted shall be inserted into a water-vaporproof bag fabricated from Class I barrier material conforming to the requirements of Military Specification MIL-B-131, and closure heat-sealed. The outer carton shall be a Class 2, Style RSC fiberboard box conforming to the requirements of Federal Specification PPP-B-636, W5c or W5s compliance symbol.

5.1.1.3 Intermediate package. - (Not applicable)

5.1.2 Level B. - (Not applicable)

5.1.3 Level C.

5.1.3.1 Cleaning, drying, and preservative application. - (Not applicable)

5.1.3.2 Unit packaging. - Unit packaging shall be in accordance with Method III of Military Specification MIL-P-116. The unit package shall contain one (1) test set.

5.1.3.3 Intermediate package. - (Not applicable)

5.2 Packing.

5.2.1 Level A. - In addition to the following, Level A packaging shall be in accordance with Military Specification MIL-P-17555.

5.2.1.1 Exterior containers. - Each test set, individually packaged as specified above for Level A protection, shall be packed in an overseas type wood-cleated plywood box or in a Class 2, Style 2, 2 1/2, 3 or 4 nailed wood box conforming to the requirements of Federal Specifications PPP-B-601 or PPP-B-621, respectively.

5.2.1.1.1 When a nailed wood box is selected to be used for shipment, box lumber shall conform to Group I, II or III woods of Federal Specification PPP-B-621. The lumber shall be of one (1) inch nominal thickness and surfaced on both sides.

5.2.2 Level B. - In addition to the following, Level B packaging shall be in accordance with Military Specification MIL-P-17555.

Source: https://assist.dla.mil -- Downloaded: 20260-19T10:16Z
Check the source to verify that this is the current version before use.

5.2.2.1 Exterior containers. - Each test set, individually packaged as specified above for Leval A protection, shall be packed in a domestic type wood-cleated plywood box or in a Class 1, Style 2, 2 1/2, 3 or 4 nailed wood box conforming to the requirements of Federal Specifications PPP-B-601 or PPP-B-621, respectively.

5.2.3 Level C.

5.2.3.1 Exterior containers. - Each test set, packaged in accordance with 5.1.3.2, shall be packed in a manner which will provide protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Special marking. - The following precautionary marking shall be stenciled or stamped on the top, one (1) side and one (1) end of the shipping container:

DELICATE INSTRUMENT
HANDLE WITH CARE

5.3.2 Normal marking. - In addition to any special marking required by the contract or order, all marking shall be in accordance with Military Standard MIL-STD-129, including serial numbers.

5.3.2.1 For detailed marking requirements, see Bureau of Ordnance Drawing Number 1476194.

6. NOTES

6.1 Ordering data. - Procurement documents shall specify the title, number, and date of this specification, and exceptions to this specification, applicable drawings, and other documents. Procurement documents shall also specify the desired level of packaging and packing. The attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-P-116, and to paragraph 6.1.1 of this document.

Source: https://assist.dla.mil -- Downloaded: 20260-19T10:16Z
Check the source to verify that this is the current version before use.

6.1.1 Criteria for use of proper level of preservation, packaging and packing shall be as follows:

For Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B. This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for 6 months or less or in domestic locations for an indefinite period.

For Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 20260-19T10:16Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

Source: https://assist.dla.mil -- Downloaded: 20261O-19T10:16Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Commander
Naval Air Systems Comand
Attn: Code 52021
Washington, D C 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:16Z
Check the source to verify that this is the current version before use.